STAT-S 650 TIME SERIES ANALYSIS HOMEWORK 1

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Theory

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \\ b_{13} & b_{23} \end{bmatrix}$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

Matrix theory

1) Prove that $(A \cdot B)' = B' \cdot A'$

From above.

$$A \cdot B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \\ b_{13} & b_{23} \end{bmatrix}$$

3×3 $\qquad$ 3×2

$$A \cdot B = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{12} + a_{13}b_{13} & a_{11}b_{21} + a_{12}b_{22} + a_{13}b_{23} \\ a_{21}b_{11} + a_{22}b_{12} + a_{23}b_{13} & a_{21}b_{21} + a_{22}b_{22} + a_{23}b_{23} \\ a_{31}b_{11} + a_{32}b_{12} + a_{33}b_{13} & a_{31}b_{21} + a_{32}b_{22} + a_{33}b_{23} \end{bmatrix}$$

$$(AB)' = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{12} + a_{13}b_{13} & a_{21}b_{11} + a_{22}b_{12} + a_{23}b_{13} & a_{31}b_{11} + a_{32}b_{12} + a_{33} \\ a_{11}b_{21} + a_{12}b_{22} + a_{13}b_{23} & a_{21}b_{21} + a_{22}b_{22} + a_{23}b_{23} & a_{31}b_{21} + a_{32}b_{22} + a_{33}b_{23} \end{bmatrix}$$

Now $B' \cdot A'$

$$\Rightarrow B' = \underset{2\times 3}{\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}} \cdot A' = \underset{3\times 3}{\begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}}$$

$$B' \cdot A' = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{12} + a_{13}b_{13} & a_{21}b_{11} + a_{22}b_{12} + a_{23}b_{13} & a_{31}b_{11} + a_{32}b_{12} + a_{31}b_{13} \\ a_{11}b_{21} + a_{12}b_{22} + a_{13}b_{23} & a_{21}b_{21} + a_{22}b_{22} + a_{23}b_{23} & a_{31}b_{21} + a_{32}b_{22} + a_{33}b_{23} \end{bmatrix}$$

Hence $(A \cdot B)' = B' \cdot A'$

2. $(AC)^{-1} = C^{-1} A^{-1}$

Assuming C^{-1} & A^{-1} exist

For C^{-1} to exits we can use Matrix definition of $M \cdot M^{-1} = I$

$M^{-1} \cdot M = I$

⇒ We need to check

$(AC)^{a} (C^{-1} \cdot A^{-1}) = I$ & $(C^{-1} \cdot A^{-1})(AC)^{a} = I$

⇒ Using the Associative property of Matrices

$(AC)^{a} (C^{-1} \cdot A^{-1}) = I$

$= A (C \cdot C^{-1}) \cdot A^{-1} = A (I) \cdot A^{-1}$

$= A \cdot A^{-1} = I$

→ $(C^{-1} A^{-1})(AC)$ ⇒ $C^{-1} \cdot (A^{-1} \cdot A) \cdot C$

$= C^{-1} \cdot (I) \cdot C$

$= C^{-1} \cdot C = I$

Since both give Identity,

$C^{-1} \cdot A^{-1}$ must be Inverse of $(A \cdot C)$

Therefor $(AC)^{-1} = C^{-1} \cdot A^{-1}$

3) Prove that $(A')^{-1} = (A^{-1})'$

Assuming Inverse exists.

From above we know that $(A \cdot B)' = B' \cdot A'$

$\Rightarrow A^{-1} \cdot A = I$

$\Rightarrow (A^{-1} \cdot A)' = I' \Rightarrow I' = I$

$\Rightarrow (A^{-1} \cdot A)' = A' \cdot (A^{-1})'$

$\Rightarrow I = A' \cdot (A^{-1})'$ Right Inverse

$\Rightarrow A \cdot A^{-1} = I$

$\Rightarrow$ Taking transpose.

$(A \cdot A^{-1})' = I' \Rightarrow (A A^{-1})' = (A^{-1})' A'$

$(A^{-1})' A' = I$

left Inverse.

Therefore $(A')^{-1} = (A^{-1})'$

4) Y is a 3x1 Matrix with

$$\mu_Y = \begin{bmatrix} \mu \\ -\mu \\ \mu \end{bmatrix} \qquad \Sigma_{YY} = \sigma^2 \cdot \underline{I}$$

ϵ is an Independent random variable with zero Mean $\mu = 0$ & Variance $= \phi^2$

$\Rightarrow \mathbb{E}(\epsilon) = 0 \quad Var(\epsilon) = \phi^2$

$$W = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \cdot Y + \epsilon$$

Need to find Mean & Variance

lets take $a = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$

$\Rightarrow$ Mean of W = $W = aY + \epsilon$

$\Rightarrow$ Taking expectation $\mathbb{E}[W] = \mathbb{E}[aY + \epsilon]$

$\Rightarrow \mathbb{E}[W] = a\,\mathbb{E}[Y] + \mathbb{E}[\epsilon]$

from above $\mathbb{E}[Y] = \mu_Y = \begin{bmatrix} \mu \\ -\mu \\ \mu \end{bmatrix}$ $\qquad \mathbb{E}(\epsilon) = 0$

• Therefor $E[W] = a\,\mu_y + 0$

$$= \underset{1\times 3}{\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}} \underset{3\times 1}{\begin{bmatrix} \mu \\ -\mu \\ \mu \end{bmatrix}}$$

$$\Rightarrow 1\cdot\mu + (-2\cdot -\mu) + 1\cdot\mu$$

$$= \mu + 2\mu + \mu$$

$$= \boxed{4\mu}$$

$$\boxed{\text{Mean} = 4\mu}$$

Variance $= W = aY + \epsilon$

$$\Rightarrow \text{Var}(W) = \text{Var}(aY + \epsilon)$$

$$= \text{Var}(ay) + \text{Var}(\epsilon)$$

$$\Rightarrow \text{Var}(aY) = a\,\Sigma_{yy}\,a'$$

Given ~~Variance~~ $\Sigma_{yy} = \sigma^2 I$

$$\text{Var}(aY) = a(\sigma^2 I)\,a' = \sigma^2\, a\, I\, a' = \sigma^2 \cdot a \cdot a'$$

$$a\cdot a' = \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = 1 + (-2\times -2) + 1$$

$$= 6$$

$$\boxed{\text{Var} = 6\sigma^2}$$

$$\text{Var}(W) = 6\sigma^2 + \phi^2$$

$\therefore$ $\mathbb{E}[W] = 4\mu$

$\text{Var}(W) = 6\sigma^2 + \phi^2$

5) Covariance b/w W & Y.

$\text{Cov}(W, Y)$

$$\text{Cov}(W, Y) = \mathbb{E}\left[(W - \mathbb{E}[W])(Y - \mathbb{E}[Y])'\right]$$

$W = aY + \epsilon$

$\text{Cov}(W, Y) = \text{Cov}(aY + \epsilon, Y)$

$\Rightarrow \text{Cov}(W, Y) = \text{Cov}(aY, Y) + \text{Cov}(\epsilon, Y)$

$\text{Cov}(\epsilon, Y) = 0$ due to Independence.

$\therefore$ $\text{Cov}(aY, Y) = a\,\text{Cov}(Y, Y) = a\,\Sigma_{YY}$

$\Sigma_{YY} = \sigma^2 I$ from above.

$$Cov(W, Y) = a \Sigma_{YY} = a(\sigma^2 I) = \sigma^2 a$$

$$Cov(W, Y) = \begin{bmatrix} \sigma^2 & -2\sigma^2 & \sigma^2 \end{bmatrix}$$

From $a = [1 \; -2 \; 1]$

6) Correlation Matrix

$$Cov(W, Y) = \sigma^2 [1 \; -2 \; 1]$$

$$Var(W) = 6\sigma^2 + \phi^2$$

$$Var(Y_1) = \sigma^2$$

Correlation Matrix between W & Y is.

$$\rho_{W,Y} = \frac{Cov(W, Y)}{\sqrt{Var(W)} \; \sqrt{Var(Y)}}$$

$Cov(W, Y) = \sigma^2$ $\quad\quad$ $\sqrt{Var(W)} = \sqrt{6\sigma^2 + \phi^2}$

$Cov(W, Y_2) = -2\sigma^2$ $\quad\quad$ $\sqrt{Var Y} = \sqrt{\sigma^2} = \sigma$

$Cov(W, Y_3) = \sigma^2$

$$\rho_{W,Y_1} = \frac{Cov(W, Y_1)}{\sqrt{6\sigma^2 + \phi^2} \cdot \sigma}$$

$$\therefore \rho_{w,v} = \left[\frac{\sigma^2}{\sigma\sqrt{6\sigma^2+\phi^2}} \quad \frac{-2\sigma^2}{\sigma\sqrt{6\sigma^2+\phi^2}} \quad \frac{\sigma^2}{\sigma\sqrt{6\sigma^2+\phi^2}} \right]$$

$$= \left[\frac{\sigma}{\sqrt{6\sigma^2+\phi^2}} \quad \frac{-2\sigma}{\sqrt{6\sigma^2+\phi^2}} \quad \frac{\sigma}{\sqrt{6\sigma^2+\phi^2}} \right]$$

$$= \sqrt{6\sigma^2+\phi^2} \quad - \quad \sigma\sqrt{6+\phi^2/\sigma^2}$$

$$\Rightarrow \left[\frac{1}{\sqrt{6+\phi^2/\sigma^2}} \quad \frac{-2}{\sqrt{6+\phi^2/\sigma^2}} \quad \frac{1}{\sqrt{6+\phi^2/\sigma^2}} \right]$$

Final Correlation Matrix.